higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PLUMBING
NQF LEVEL 3

(12030373)

21 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages and 1 diagram sheet.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use a pen with only BLUE or BLACK ink.
6. Write neatly and legibly.

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QUESTION 1: COLD WATER RETICULATION

1.1 Name FOUR factors that will affect the rate of the flow of water in a water supply system. (4)

1.2 What guidelines should be followed when laying a water supply pipe near a drain pipe? (2)

1.3 What is the *flow rate demand*? (2)

1.4 Which method can be used to prevent movements of a water supply pipe that can potentially damage fittings and cause the pipe to start leaking, when the installation is:

1.4.1 Underground

1.4.2 Above the ground

(2 × 1) (2)

1.5 What is a working drawing that can be used when calculating the quantities of materials for the installation of residential cold water systems? (4)

1.6 When opening a tap in an installation, water hammer or vibration noises may occur.

List FOUR common causes of these conditions. (4)

1.7 Water intended for human consumption must be of an acceptable quality or standard.

Name TWO general quality requirements for potable water. (2)

[20]

QUESTION 2: HOT WATER RETICULATION

2.1 What is the best type of joint to use for connecting valves and terminal fittings to the following?

2.1.1 Copper pipes in a hot water supply system

2.1.2 GMS pipes in a hot water supply system

(2 × 1) (2)

2.2 Discuss the importance of SANS regulations when installing hot water reticulation and geysers. (4)

2.3 Describe the working principles of a typical pressure control valve. (5)

2.4 Briefly explain why thermal insulation is used on hot water pipework. (5)

2.5 State FOUR disadvantages of a combination geyser. (4)

[20]

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QUESTION 3: ABOVE-GROUND DRAINAGE SYSTEMS

3.1 What are the requirements for materials used in the installation of soil, waste and vent systems? (3)

3.2 What should be used to secure pipes to a wall? (2)

3.3 Write out the following sanitary fittings and fixtures abbreviations in full:

3.3.1 WC

3.3.2 BT

3.3.3 SW

3.3.4 U

3.3.5 WHB

(5 × 1) (5)

3.4 Describe the principles of a single-stack system. (3)

3.5 State TWO disadvantages of the two-pipe system. (2)

[15]

QUESTION 4: SANITARY INSTALLATION

4.1 What are splash backs used for and what materials can they be made of? (4)

4.2 Make a drawing of a vertical section through a wash-down water closet (P-trap). (6)

4.3 What kind of water seal trap is fitted into a bath and how should it be connected? (3)

4.4 In which way is the water supply usually connected to a wall-mounted, single-bowl urinal? (2)

[15]

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QUESTION 5: RAINWATER SYSTEM

5.1 List THREE different materials from which gutters are manufactured. (3)

5.2 Describe THREE ways in which GMS gutters and fittings can be joined. (3)

5.3 Explain the function of the following:

5.3.1 Gutter stop

5.3.2 Spreader pipe

5.3.3 Downpipe shoe

(3 × 2) (6)

5.4 Where is the water collected that is discharged by the rainwater system? (1)

5.5 Explain the function of the following:

5.5.1 Gully

5.5.2 Grease trap

(2 × 1) (2)

[15]

QUESTION 6: VENT-PIPE FLUSHING

In FIGURE 1 on the DIAGRAM SHEET (attached), TWO views of a square pipe are shown.

Using the parallel-line method to develop the full pattern on a scale 1:5 of the truncated pipe. **[10]**

Do NOT make any allowances for the seam.

QUESTION 7: TRENCHING, EXCAVATION AND TRENCH SUPPORT

7.1 Explain what is meant by *excavating trenches to line and level*. (4)

7.2 What are the most effective gradients for a 100-mm-diameter drain pipe? (1)

[5]

TOTAL: 100

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DIAGRAM SHEET

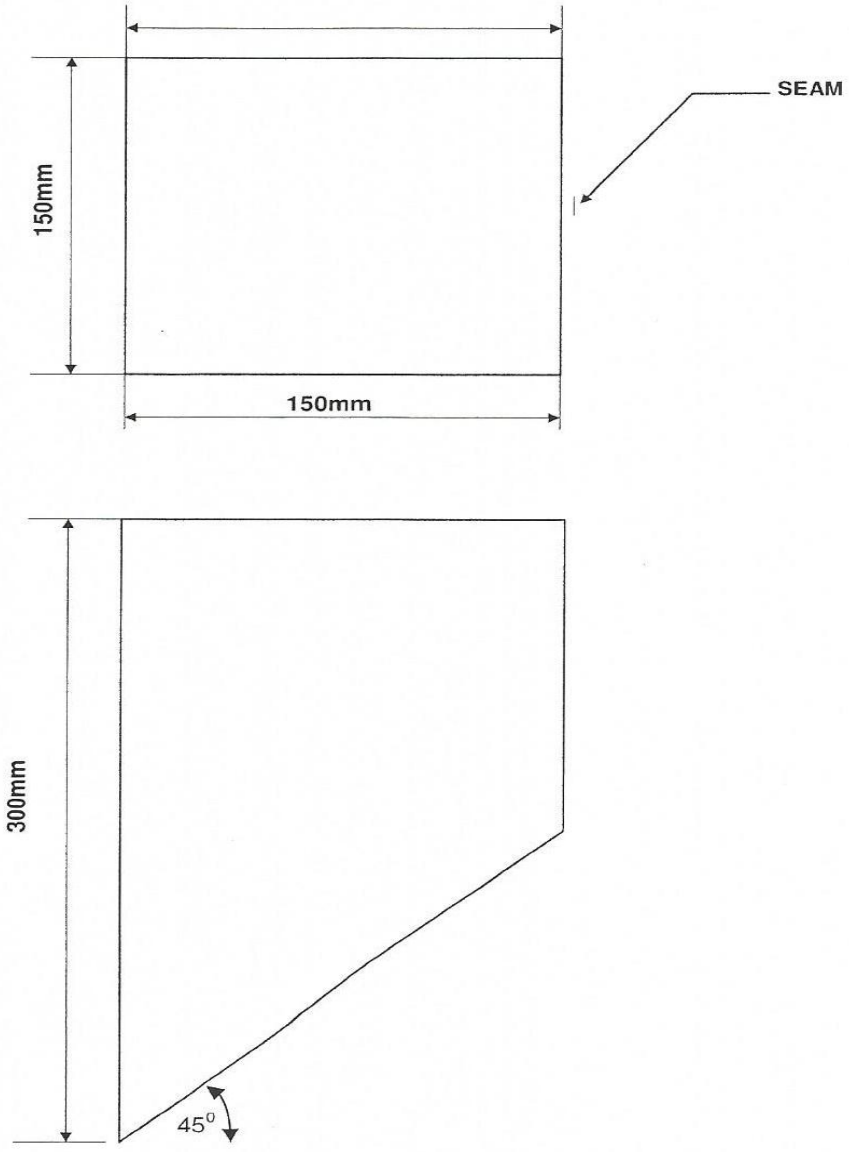


FIGURE 1

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